

Smart Campus Management System

UML Class Diagram Documentation

Project	Smart Campus Management System
Language	C++ (OOP)
Modules	Person, Course, Enrollment, Finance, Hostel, Library
Diagram Type	UML Class Diagram
Pages	6 diagram pages + cover

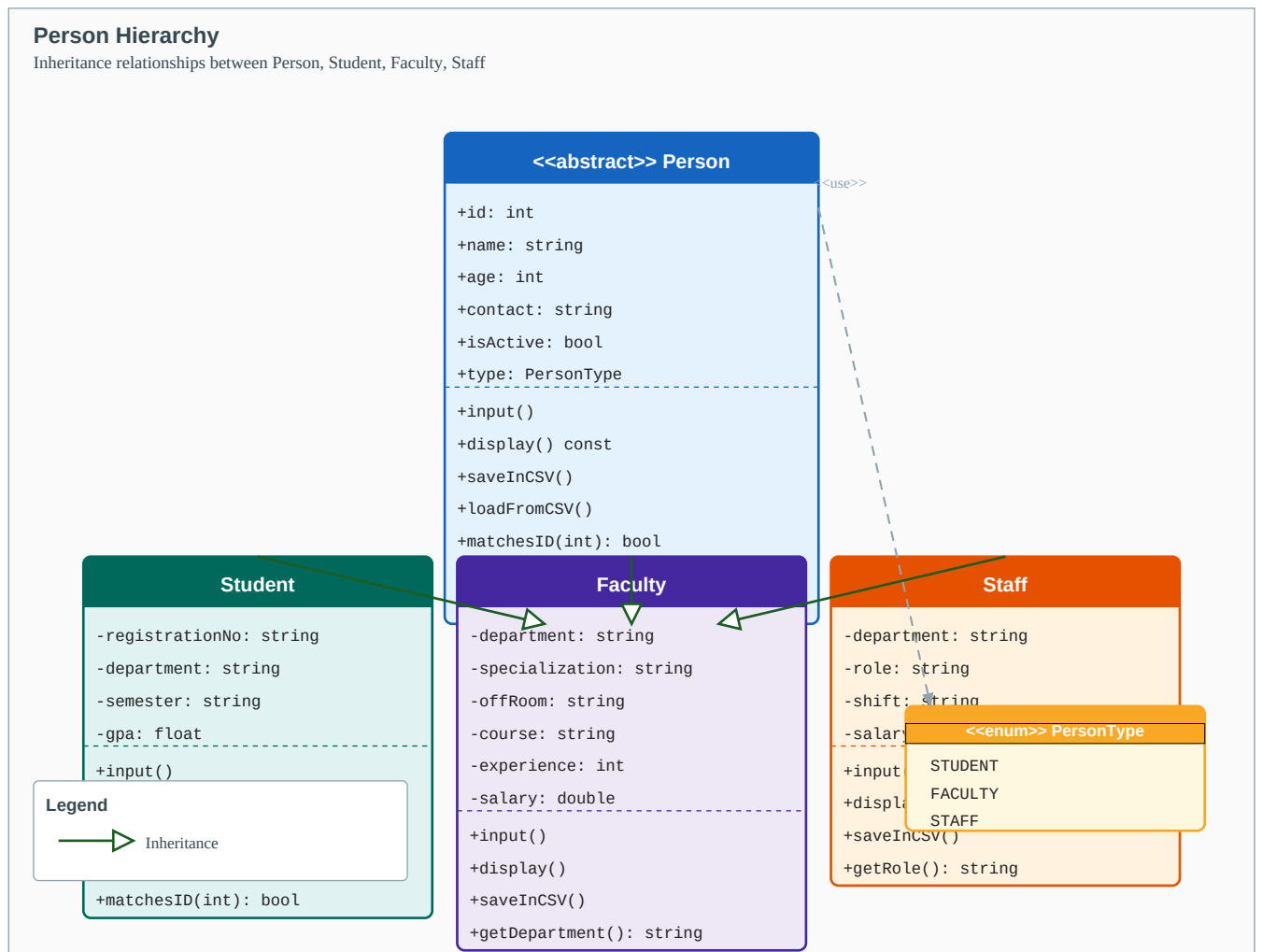
Module Overview

This document contains the complete UML class diagram for the Smart Campus Management System built in C++. The system uses object-oriented principles including inheritance, composition, and encapsulation across 6 core modules.

Module	Classes	Description
Person	Person, Student, Faculty, Staff	Manages all campus members with inheritance
Finance	FeeRecord, Invoice	Handles fee tracking and invoice generation
Hostel	Room, HostelBlock, HostelManager	Manages hostel rooms and blocks
Library	LibraryItem, Book, Journal, Library	Tracks books, journals and library items
Academic	Course, Enrollment	Course catalog and student enrollments
Reporting	Reports	Static class generating all system reports

1. Person Hierarchy

Inheritance tree: Person → Student, Faculty, Staff

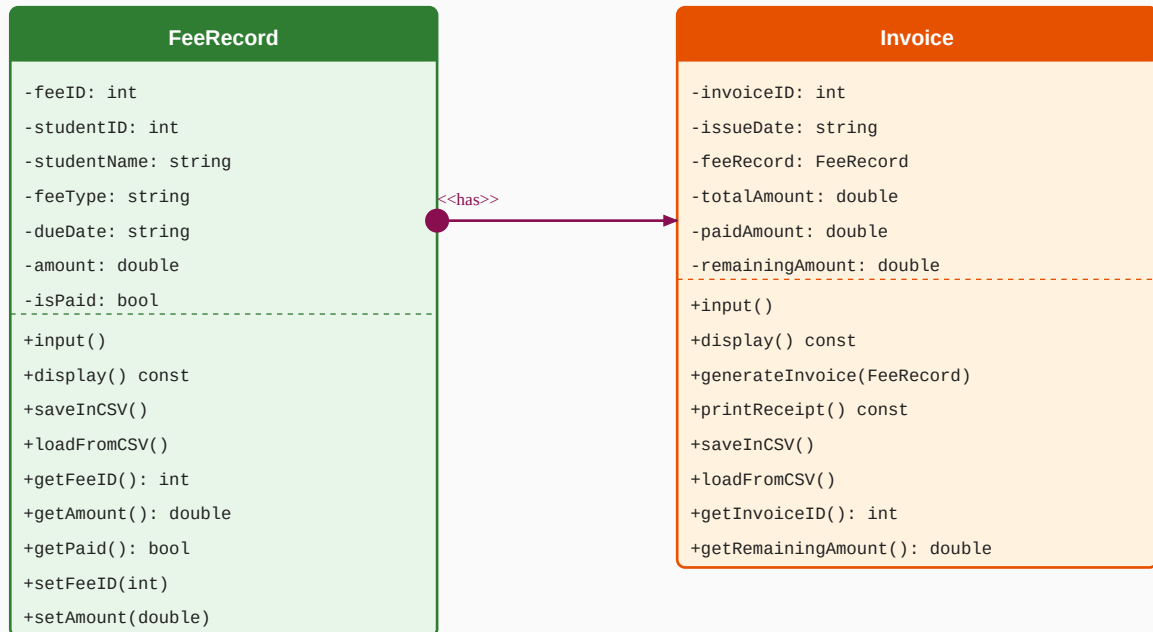


2. Finance Module

FeeRecord has-a composition with Invoice

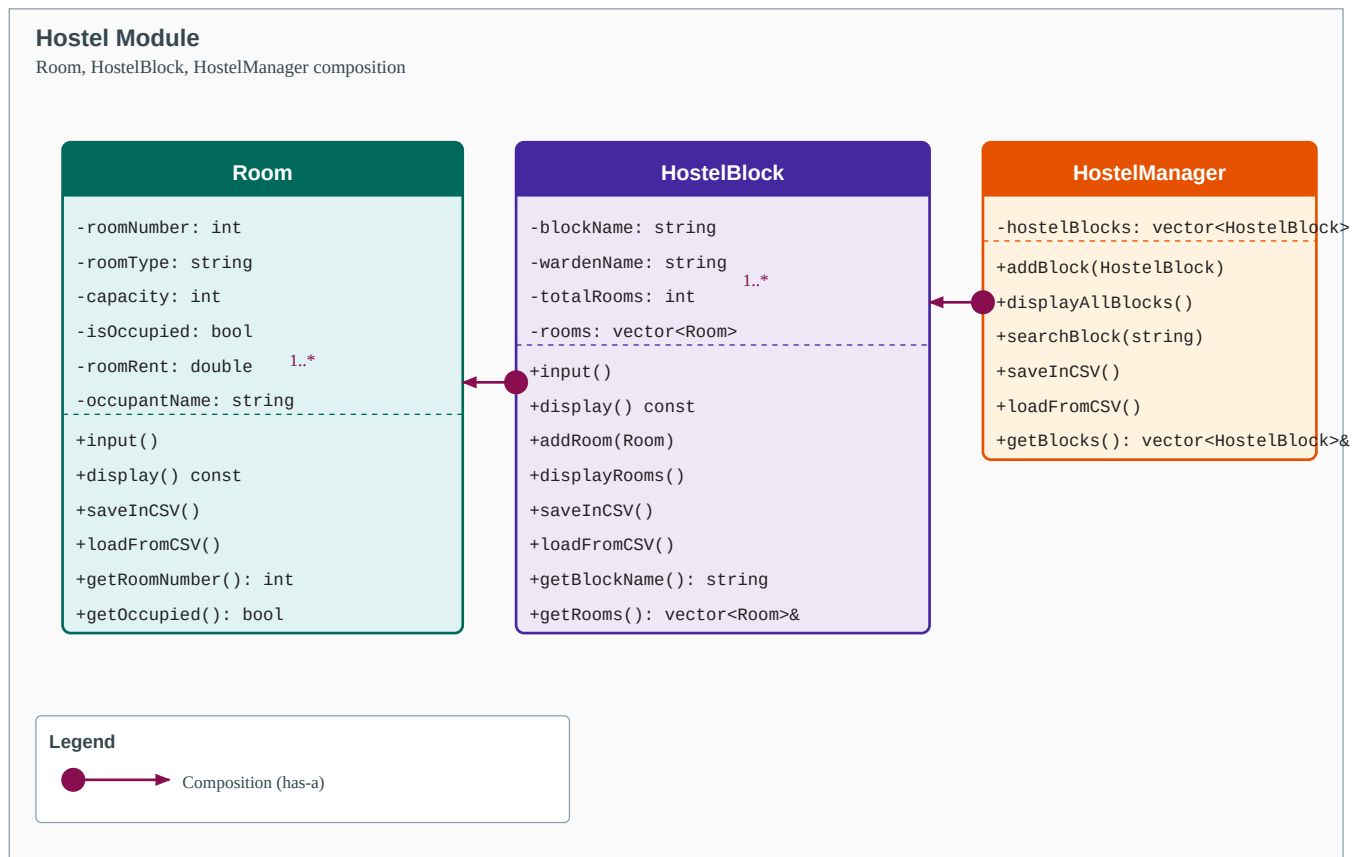
Finance Module

FeeRecord and Invoice relationship



3. Hostel Module

Room → HostelBlock → HostelManager composition chain

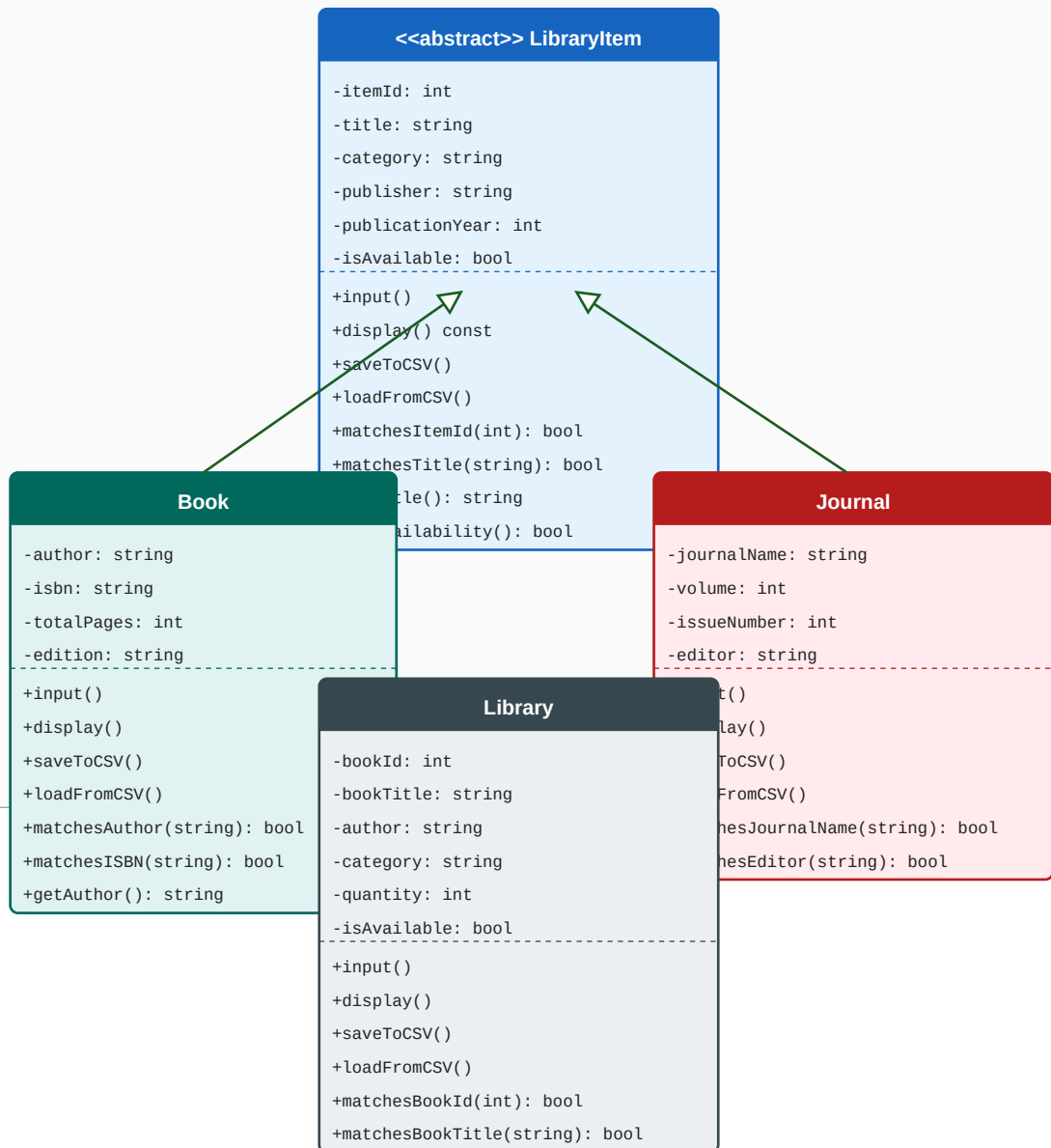


4. Library Module

LibraryItem → Book, Journal inheritance; standalone Library

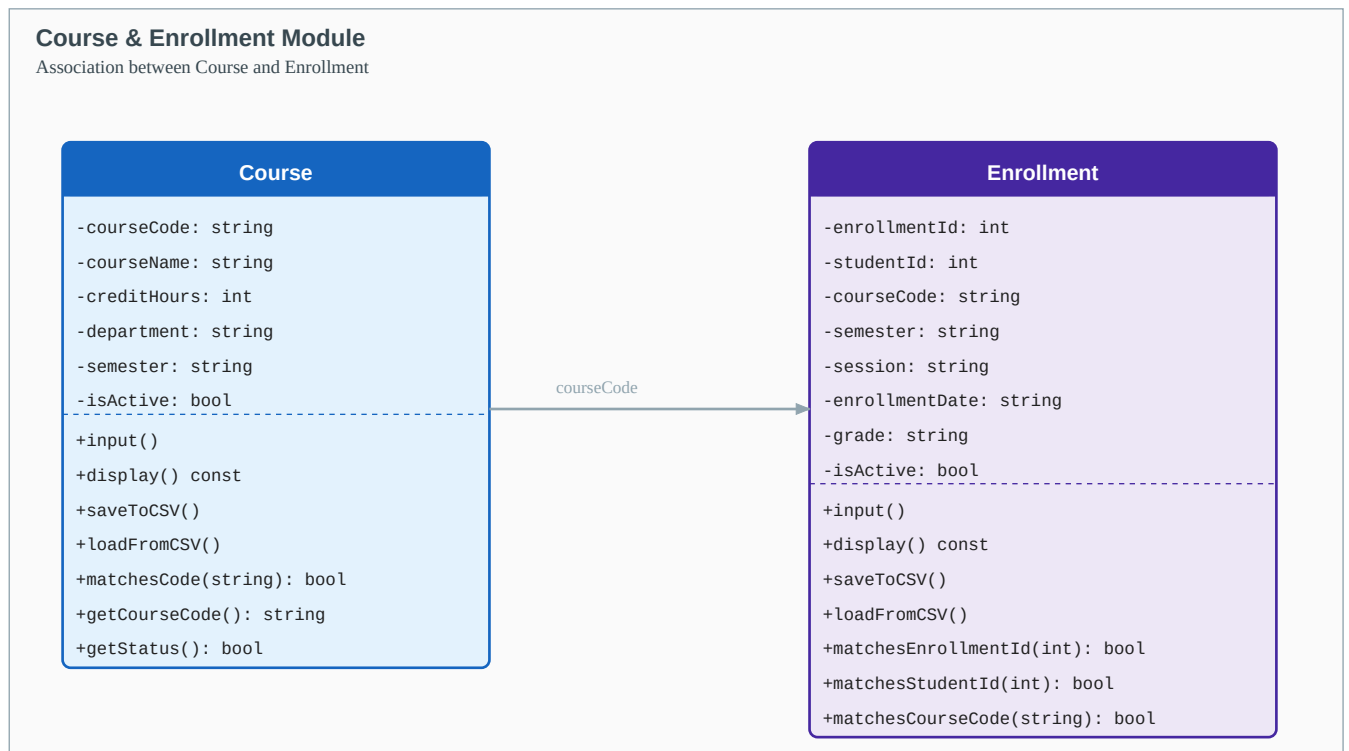
Library Module

LibraryItem base class with Book and Journal subclasses



5. Course & Enrollment

Course associated with Enrollment via courseCode



6. System Overview

main.cpp orchestrates all modules with Reports and CSV storage

